

Network Technologies

Complete Study Notes

Computer Networks • Data Communication • Internet Architecture

1. Overview of Network Types

Computer networks are classified by their geographic coverage, purpose, and scale. Understanding these categories is fundamental to network design and administration.

Network Type	Full Name	Coverage	Speed Range	Typical Use
PAN	Personal Area Network	~10 meters	1 Mbps – 480 Mbps	Personal device connectivity
LAN	Local Area Network	Up to 1 km	10 Mbps – 10 Gbps	Office/home networking
SAN	Storage Area Network	Building/campus	1 Gbps – 128 Gbps	High-speed storage access
MAN	Metropolitan Area Network	City-wide (50 km)	10 Mbps – 10 Gbps	City infrastructure, ISPs
WAN	Wide Area Network	Countries/Global	Varies	Internet, enterprise links

2. Personal Area Networks (PANs)

A Personal Area Network (PAN) is the smallest and most personal type of network, typically used to connect devices within an individual's immediate environment, usually within a range of about 10 meters.

2.1 Key Characteristics

- Coverage range: Typically within 10 meters (short-range)
- Designed for personal device interconnection
- Low power consumption, especially wireless PANs
- Usually centered around a single individual
- Can be wired (USB) or wireless (Bluetooth, infrared)

2.2 Technologies Used in PANs

Bluetooth

- IEEE standard: 802.15.1
- Range: ~10 meters (Class 2) up to 100 meters (Class 1)
- Frequency: 2.4 GHz ISM band
- Data rate: Up to 3 Mbps (Classic); up to 50 Mbps (Bluetooth 5.x)
- Uses frequency hopping spread spectrum (FHSS) to avoid interference
- Common uses: Wireless headsets, keyboards, mice, smartwatches, speakers

Infrared (IrDA)

- Requires line-of-sight communication
- Range: ~1 meter
- Speed: Up to 16 Mbps
- Largely replaced by Bluetooth but still used in remote controls

Zigbee (IEEE 802.15.4)

- Very low power, low data rate
- Range: 10–100 meters
- Data rate: 250 Kbps
- Common in IoT devices, smart home automation, medical sensors

USB / FireWire (Wired PAN)

- USB 3.0: Up to 5 Gbps; USB 4.0: Up to 40 Gbps
- Primarily for device-to-device or device-to-host connections

2.3 PAN Applications

- Connecting smartphones to wireless earbuds or headsets
- Transferring files between nearby devices
- Wearable health monitors sending data to phones
- Smart home device control (thermostats, locks, lights)
- Medical body area networks (BANs) for patient monitoring

 *BANs (Body Area Networks) are a subcategory of PANs used specifically for health/medical monitoring devices worn on or inside the body.*

3. Storage Area Networks (SANs)

A Storage Area Network (SAN) is a high-speed, dedicated network that provides block-level access to data storage. Unlike traditional networks, a SAN is specifically designed to handle storage traffic separately from regular LAN traffic.

3.1 Key Characteristics

- Operates at block level (not file level like NAS)
- Dedicated, high-bandwidth network for storage traffic

- Provides high availability and redundancy
- Low latency, critical for databases and enterprise applications
- Typically confined to a data center or campus

3.2 SAN vs NAS Comparison

Feature	SAN	NAS
Access Level	Block-level	File-level
Protocol	Fibre Channel, iSCSI	NFS, SMB/CIFS
Performance	Very High	Moderate
Cost	High	Lower
Scalability	High	Moderate
Typical Use	Databases, VMs	File sharing, backups

3.3 SAN Protocols and Technologies

Fibre Channel (FC)

- Traditional SAN protocol, very high performance
- Speeds: 1, 2, 4, 8, 16, 32, 64 Gbps
- Uses dedicated fibre channel switches and HBAs (Host Bus Adapters)
- Highly reliable with built-in error detection

iSCSI (Internet SCSI)

- Encapsulates SCSI commands over TCP/IP networks
- Allows SAN over standard Ethernet infrastructure
- More cost-effective than Fibre Channel
- Widely used in SME environments

Fibre Channel over Ethernet (FCoE)

- Combines Fibre Channel and Ethernet into a single network
- Reduces cabling and hardware costs
- Requires Data Center Bridging (DCB) capable switches

NVMe over Fabrics (NVMe-oF)

- Next-generation protocol for flash storage
- Extremely low latency (microseconds)
- Works over FC, Ethernet (RDMA), and TCP

3.4 SAN Architecture Components

- Host Bus Adapters (HBAs): Connect servers to SAN fabric
- SAN Switches: Dedicated switches routing storage traffic
- Storage Arrays: RAID-based disk arrays providing storage
- SAN Management Software: Zoning, LUN masking, monitoring
- Zoning: Logical partitioning of the SAN fabric for security and management

4. Local Area Networks (LANs)

A Local Area Network (LAN) connects computers and devices within a limited geographic area such as a home, office, school, or building. It is the most common type of network in everyday use.

4.1 Key Characteristics

- Coverage: Typically within a single building or campus (up to ~1 km)
- High data transfer rates (10 Mbps to 10 Gbps and beyond)
- Low latency due to short distances
- Usually privately owned and managed
- Supports both wired (Ethernet) and wireless (Wi-Fi) connections

4.2 Ethernet (IEEE 802.3) – Wired LAN

Common Ethernet Standards

Standard	Speed	Cable	Max Distance
10BASE-T	10 Mbps	Cat3 UTP	100 m
100BASE-TX (Fast Ethernet)	100 Mbps	Cat5 UTP	100 m
1000BASE-T (Gigabit Ethernet)	1 Gbps	Cat5e/Cat6	100 m
10GBASE-T	10 Gbps	Cat6a/Cat7	100 m
40/100GBASE	40/100 Gbps	Fibre Optic	Varies

Ethernet Frame Structure

- Preamble (7 bytes): Synchronization sequence
- Start Frame Delimiter - SFD (1 byte): Marks start of frame
- Destination MAC Address (6 bytes)
- Source MAC Address (6 bytes)
- EtherType/Length (2 bytes): Protocol type or data length
- Data/Payload (46–1500 bytes)
- Frame Check Sequence - FCS (4 bytes): CRC error detection

 Maximum Transmission Unit (MTU) for standard Ethernet is 1500 bytes. Jumbo frames allow up to 9000 bytes.

4.3 Wireless LAN – Wi-Fi (IEEE 802.11)

Standard	Frequency	Max Speed	Range	Year
802.11b	2.4 GHz	11 Mbps	~35 m indoor	1999
802.11g	2.4 GHz	54 Mbps	~38 m indoor	2003
802.11n (Wi-Fi 4)	2.4/5 GHz	600 Mbps	~70 m indoor	2009
802.11ac (Wi-Fi 5)	5 GHz	3.5 Gbps	~35 m indoor	2013
802.11ax (Wi-Fi 6/6E)	2.4/5/6 GHz	9.6 Gbps	~30 m indoor	2019
802.11be (Wi-Fi 7)	2.4/5/6 GHz	46 Gbps	~30 m indoor	2024

4.4 LAN Topologies

- Bus Topology: All devices share a single communication line; obsolete
- Star Topology: All devices connect to a central switch/hub; most common
- Ring Topology: Devices form a closed ring; used in Token Ring (obsolete)
- Mesh Topology: Every device connects to every other device; high redundancy
- Hybrid Topology: Combination of two or more topologies

4.5 LAN Devices

- **Hub:** Layer 1 device; broadcasts all traffic to all ports; obsolete
- **Switch:** Layer 2 device; forwards frames based on MAC addresses; creates collision domains
- **Router:** Layer 3 device; routes packets between networks based on IP addresses
- **Access Point (AP):** Provides wireless connectivity; bridges wireless clients to wired LAN
- **Network Interface Card (NIC):** Hardware enabling devices to connect to LAN

5. Metropolitan Area Networks (MANs)

A Metropolitan Area Network (MAN) spans a city or a large campus — larger than a LAN but smaller than a WAN. It provides connectivity across a metropolitan area, typically covering distances up to 50 km.

5.1 Key Characteristics

- Coverage: Typically 5–50 km (city-wide)
- Data rates: 10 Mbps to 10+ Gbps
- Often operated by ISPs, telecom companies, or municipal authorities
- Serves as a backbone connecting multiple LANs within a city
- Can use fibre optic, microwave, or other high-speed technologies

5.2 MAN Technologies

SONET/SDH (Synchronous Optical Networking)

- High-speed optical networking standard
- Speeds: OC-1 (51.84 Mbps) to OC-768 (39.8 Gbps)
- Highly reliable with automatic protection switching
- Used by telecommunications providers

Metro Ethernet

- Extends Ethernet technology to MAN distances
- Uses fibre optic cables for high-speed connectivity
- Based on familiar Ethernet protocols, easier to manage
- Supports VLAN tagging (IEEE 802.1Q) for traffic separation

WiMAX (IEEE 802.16)

- Wireless MAN technology
- Range: Up to 50 km (line of sight)
- Speed: Up to 1 Gbps (802.16m)
- Provides last-mile broadband wireless connectivity

FDDI (Fibre Distributed Data Interface)

- Older MAN technology using a dual token ring
- Speed: 100 Mbps
- Highly fault-tolerant due to redundant ring
- Largely replaced by Gigabit Ethernet

5.3 MAN Applications

- Connecting government offices across a city
- City-wide broadband internet services
- Cable television networks
- University campus networks spanning multiple buildings
- Municipal surveillance and traffic management systems

6. Wide Area Networks (WANs)

A Wide Area Network (WAN) spans large geographic areas — countries, continents, or the entire globe. The Internet is the most well-known WAN. WANs connect multiple LANs and MANs together.

6.1 Key Characteristics

- Coverage: Country-wide to global
- Usually operated by telecommunications providers (ISPs, carriers)
- Higher latency compared to LAN and MAN
- Uses leased lines, satellite, or public infrastructure
- Lower speeds and higher costs per unit of data compared to LANs

6.2 WAN Technologies

Leased Lines

- Dedicated point-to-point connections between two locations
- T1 (1.544 Mbps), T3 (44.736 Mbps), E1 (2.048 Mbps)
- Consistent bandwidth, high cost
- Used by enterprises for guaranteed connectivity

MPLS (Multiprotocol Label Switching)

- Directs data using short path labels rather than IP addresses
- Provides QoS (Quality of Service) guarantees
- Faster forwarding as labels are processed at Layer 2.5
- Widely used by ISPs and enterprise WANs

Frame Relay (Legacy)

- Packet-switched WAN technology, now largely deprecated
- Used virtual circuits (PVCs) to connect sites
- Operated at Layer 2 of OSI model

ATM (Asynchronous Transfer Mode) – Legacy

- Fixed 53-byte cells for predictable performance
- Supported voice, video, and data over same network
- Largely replaced by IP/MPLS

SD-WAN (Software-Defined WAN)

- Modern WAN architecture using software control
- Can use multiple connection types (MPLS, broadband, LTE)
- Centralized management and policy control
- Reduces cost while improving performance and flexibility
- Increasing adoption in enterprise environments

Satellite WAN

- Provides connectivity to remote or rural areas
- High latency (~600 ms for geostationary satellites)

- LEO satellites (e.g., Starlink) offer much lower latency (~20–40 ms)
- Bandwidth is limited and shared among users

6.3 WAN Performance Metrics

- **Bandwidth:** Maximum data transfer rate (Mbps, Gbps)
- **Latency:** Time for data to travel from source to destination (ms)
- **Jitter:** Variation in latency; critical for real-time applications
- **Packet Loss:** Percentage of packets that fail to arrive
- **Throughput:** Actual data transfer rate achieved in practice

7. Network Interconnection and Relaying

Networks are connected using different devices that operate at various layers of the OSI model. The layer at which interconnection occurs determines what the device can do and how it handles network traffic.

8. Data Link Layer – Bridging and Switching

Devices operating at Layer 2 (Data Link Layer) use MAC addresses to forward frames within the same network or between connected network segments.

8.1 Bridges

- Connect two or more LAN segments at the Data Link Layer
- Use MAC address tables to forward or filter frames
- Reduce collision domains by separating network segments
- Transparent to higher-layer protocols
- Two main types: Transparent Bridges and Source Route Bridges

Transparent Bridge Operation

- **Learning:** Records source MAC addresses of incoming frames
- **Forwarding:** Sends frame to the correct port based on destination MAC
- **Filtering:** Drops frame if source and destination are on the same segment
- **Flooding:** Sends frame to all ports if destination MAC unknown

 *Spanning Tree Protocol (STP – IEEE 802.1D) prevents loops in bridged networks by blocking redundant paths while maintaining a tree topology.*

8.2 Switches

A switch is essentially a multi-port bridge that operates at wire speed. Switches have largely replaced hubs in modern LAN environments.

Key Features of Switches

- Full-duplex operation – eliminates collisions
- High port density (24, 48 ports and beyond)
- Hardware-based MAC address lookup using CAM tables
- Support for VLANs (IEEE 802.1Q) for logical network segmentation
- QoS capabilities for traffic prioritization
- Port security to limit MAC addresses per port

Types of Switches

- **Unmanaged Switch:** Plug-and-play, no configuration, basic home/SME use
- **Managed Switch:** Full configuration, VLAN support, SNMP monitoring, enterprise use
- **Layer 3 Switch:** Combines switching and routing; can forward packets based on IP addresses
- **PoE Switch:** Provides power over Ethernet to connected devices (IP phones, APs, cameras)

VLAN (Virtual LAN)

- Logically partitions a physical LAN into multiple virtual networks
- VLAN traffic is tagged using 802.1Q headers
- Devices in different VLANs cannot communicate without a router or Layer 3 switch
- Benefits: Security, broadcast traffic reduction, simplified management

Spanning Tree Protocol (STP)

- Prevents loops in switched networks with redundant links
- Elects a Root Bridge; all other switches build shortest path tree
- Blocks redundant ports to prevent loops
- Versions: STP (802.1D), RSTP (802.1w – Rapid), MSTP (802.1s – Multiple)

9. Network Layer – Routing and Switching

Layer 3 (Network Layer) devices use IP addresses to route packets between different networks. The primary device at this layer is the router.

9.1 Routers

- Operate at OSI Layer 3 (Network Layer)
- Make forwarding decisions based on destination IP addresses
- Maintain routing tables mapping networks to next-hop addresses
- Connect different networks (LAN to WAN, LAN to Internet)
- Provide NAT (Network Address Translation) for private IP address spaces

- Support access control lists (ACLs) for basic security

9.2 Routing Algorithms and Protocols

Static Routing

- Routes manually configured by an administrator
- Efficient, predictable, no overhead
- Does not adapt to network failures automatically
- Suitable for small, simple networks

Dynamic Routing Protocols

Protocol	Type	Algorithm	Use Case
RIP	IGP	Distance Vector (Bellman-Ford)	Small networks
OSPF	IGP	Link State (Dijkstra)	Large enterprise networks
EIGRP	IGP	Advanced Distance Vector	Cisco networks
BGP	EGP	Path Vector	Internet routing between ISPs
IS-IS	IGP	Link State	ISP backbone networks

Routing Table

- Contains: Destination network, subnet mask, next-hop address, metric, interface
- Router selects best path based on longest prefix match
- Administrative Distance determines protocol preference when multiple routes exist

9.3 Layer 3 Switching

- Combines routing and switching in a single device
- Uses ASICs for high-speed hardware-based routing
- Much faster than traditional software-based routers
- Common in enterprise campus networks
- Supports inter-VLAN routing without a dedicated router

9.4 IP Addressing

IPv4

- 32-bit addresses; ~4.3 billion unique addresses
- Expressed in dotted-decimal notation (e.g., 192.168.1.1)
- Classes: A (0.0.0.0–127.255.255.255), B (128–191), C (192–223)

- CIDR (Classless Inter-Domain Routing) for flexible subnetting

IPv6

- 128-bit addresses; virtually unlimited address space
- Expressed in hexadecimal colon notation (e.g., 2001:0db8::1)
- Built-in security (IPSec), auto-configuration (SLAAC)
- No broadcast; uses multicast and anycast

10. Transport Layer – Gatewaying

The Transport Layer (Layer 4) provides end-to-end communication, flow control, error detection, and multiplexing. Gateways operating at this layer translate between different transport protocols.

10.1 Transport Protocols

TCP (Transmission Control Protocol)

- Connection-oriented: establishes connection via 3-way handshake (SYN, SYN-ACK, ACK)
- Reliable delivery with acknowledgements and retransmission
- Flow control using sliding window mechanism
- Congestion control (slow start, congestion avoidance, fast retransmit)
- In-order delivery guaranteed
- Used by: HTTP/S, FTP, SMTP, SSH, Telnet

UDP (User Datagram Protocol)

- Connectionless; no handshake required
- Unreliable; no acknowledgements or retransmission
- Low overhead, low latency
- Suitable for real-time applications where speed > reliability
- Used by: DNS, DHCP, VoIP, video streaming, online gaming

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Reliable (ACK, retransmit)	Unreliable
Ordering	In-order delivery	No ordering guarantee
Speed	Slower (more overhead)	Faster (less overhead)
Header Size	20–60 bytes	8 bytes
Use Cases	Web, email, file transfer	VoIP, streaming, DNS

10.2 Transport Layer Gateways

A transport layer gateway (also called a circuit-level gateway) connects networks that use different transport protocols or provide proxy services at Layer 4.

- Translates between different transport protocol formats
- Circuit-level gateways maintain session state information
- Provide security by hiding internal network structure
- SSL/TLS offloading gateways decrypt traffic at the gateway
- Load balancers operate at Layer 4, distributing connections across servers

10.3 Port Numbers

- Ports 0–1023: Well-known ports (HTTP: 80, HTTPS: 443, FTP: 21, SSH: 22, SMTP: 25, DNS: 53)
- Ports 1024–49151: Registered ports for specific applications
- Ports 49152–65535: Dynamic/ephemeral ports for temporary client connections

11. Application Layer – Relaying and Gatewaying

Application layer gateways operate at the highest layer of the OSI model (Layer 7). They understand application-level protocols and can perform deep inspection, content transformation, and protocol conversion.

11.1 Application Layer Gateways (ALGs)

- Operate at the Application Layer (Layer 7)
- Understand application-specific protocols (HTTP, FTP, SIP, DNS)
- Can inspect, modify, filter, and translate application data
- Provide the highest level of security and granular control
- Often built into firewalls and proxy servers

11.2 Types of Application Layer Relaying

Proxy Servers

- Forward client requests to destination servers
- Can cache content to reduce bandwidth and improve performance
- Provide anonymity and content filtering
- Types: Forward Proxy (client-side), Reverse Proxy (server-side)

Reverse Proxy

- Sits in front of web servers, distributing client requests
- Provides load balancing, SSL termination, compression
- Hides backend server architecture from clients
- Examples: Nginx, HAProxy, Apache, AWS ALB

Application Firewall (WAF)

- Filters HTTP/S traffic to web applications
- Protects against: SQL injection, XSS, CSRF, DDoS
- Inspects actual application content, not just headers

API Gateways

- Manage, route, and secure API traffic
- Provide rate limiting, authentication, transformation
- Examples: Kong, AWS API Gateway, Apigee

11.3 Common Application Layer Protocols

Protocol	Port	Layer	Purpose
HTTP	80	Application	Web page transfer
HTTPS	443	Application	Secure web (TLS encrypted)
FTP	20/21	Application	File transfer
SMTP	25/587	Application	Email sending
IMAP/POP3	143/110	Application	Email retrieval
DNS	53	Application	Domain name resolution
DHCP	67/68	Application	IP address assignment
SSH	22	Application	Secure remote shell
SNMP	161	Application	Network management
LDAP	389	Application	Directory services

12. The Internet

The Internet is the world's largest Wide Area Network — a global system of interconnected computer networks using the TCP/IP protocol suite to communicate.

12.1 Internet Architecture

- Hierarchical structure: Tier 1, Tier 2, and Tier 3 ISPs
- Tier 1 ISPs: Global backbone carriers (AT&T, NTT, Telia) — peering agreements only
- Tier 2 ISPs: Regional carriers; peer with Tier 1 and purchase transit
- Tier 3 ISPs: Local ISPs; purchase transit from Tier 1/2 for end users

12.2 Key Internet Protocols

TCP/IP Protocol Suite

- Internet Protocol (IP): Routing and addressing
- TCP/UDP: Transport layer reliability and speed
- DNS: Domain name to IP address resolution
- BGP: Inter-domain routing between ISPs
- HTTP/HTTPS: Web content delivery

Internet Exchange Points (IXPs)

- Physical locations where different networks interconnect
- Reduce costs and improve performance through direct peering
- Major IXPs: AMS-IX (Amsterdam), LINX (London), DE-CIX (Frankfurt)

12.3 Internet Addressing and DNS

- ICANN (Internet Corporation for Assigned Names and Numbers) manages domain names and IP allocation
- IANA delegates IP blocks to Regional Internet Registries (RIRs): ARIN, RIPE, APNIC, AFRINIC, LACNIC
- DNS hierarchy: Root servers → TLD servers (.com, .org) → Authoritative Name Servers → Resolvers
- IPv4 exhaustion led to IPv6 adoption and widespread use of NAT

12.4 Internet Access Technologies

Technology	Speed	Medium	Typical Use
DSL (ADSL/VDSL)	1–100 Mbps	Copper phone line	Home broadband
Cable (DOCSIS 3.1)	Up to 10 Gbps	Coaxial cable	Home/business broadband
Fibre (FTTH/FTTP)	100 Mbps – 10 Gbps	Optical fibre	Home/enterprise
4G/LTE	50–150 Mbps	Radio	Mobile broadband
5G	Up to 10 Gbps	Radio (mmWave/sub-6G Hz)	Mobile, fixed wireless
Satellite (LEO)	50–500 Mbps	Satellite	Rural/remote areas

12.5 Internet Security Fundamentals

- Firewalls: Filter traffic at network/transport/application layers
- SSL/TLS: Encryption for web and application traffic (HTTPS)
- VPN: Encrypted tunnel over public internet for private communication

- IDS/IPS: Intrusion Detection/Prevention Systems monitor for attacks
- DDoS Mitigation: Traffic scrubbing, rate limiting, Anycast routing

13. OSI Model Reference

Understanding which device or protocol operates at each layer is fundamental to network troubleshooting and design.

Layer	Name	Protocols/Standards	Devices/Functions
7	Application	HTTP, FTP, SMTP, DNS, DHCP	Application Gateways, Proxies, ALGs
6	Presentation	SSL/TLS, JPEG, MPEG, ASCII	Encryption/Compression
5	Session	NetBIOS, RPC, SQL	Session management
4	Transport	TCP, UDP, SCTP	Transport Gateways, Load Balancers
3	Network	IP, ICMP, ARP, BGP, OSPF	Routers, Layer 3 Switches
2	Data Link	Ethernet, Wi-Fi, PPP, Frame Relay	Switches, Bridges, APs
1	Physical	RS-232, USB, Ethernet cabling	Hubs, Repeaters, NICs, Cables

14. Key Acronyms and Terms

Acronym	Full Form
PAN	Personal Area Network
LAN	Local Area Network
MAN	Metropolitan Area Network
WAN	Wide Area Network
SAN	Storage Area Network
NAS	Network Attached Storage
MAC	Media Access Control
IP	Internet Protocol
TCP	Transmission Control Protocol
UDP	User Datagram Protocol

Acronym	Full Form
DNS	Domain Name System
DHCP	Dynamic Host Configuration Protocol
NAT	Network Address Translation
VPN	Virtual Private Network
QoS	Quality of Service
VLAN	Virtual Local Area Network
ISP	Internet Service Provider
BGP	Border Gateway Protocol
OSPF	Open Shortest Path First
MPLS	Multiprotocol Label Switching
CSMA/CD	Carrier Sense Multiple Access / Collision Detection
ALG	Application Layer Gateway
IXP	Internet Exchange Point
ICANN	Internet Corporation for Assigned Names and Numbers